

Patient Monitoring System (PMS)

Part 1: instructions

In part 1 of the assignment, you will identify and prioritize **Architecturally-Significant Requirements** (ASRs) and document the most important ones as **Quality Attribute Scenarios** (QASes).

1 Identification of ASRs and Utility Tree

- Study the description of the application case carefully.
- Document 15 of the most important architecturally-significant requirements that will affect the development of the described system. (Refer to the lecture for some pointers and strategies on how to identify the relevant requirements.)
- Provide a short description or summary paragraph (1–2 sentences per identified ASR) about the nature of the requirement and the constraints it imposes on the system. Make sure to including a measurable or verifiable condition that allows assessing when the requirement is met.
- Per ASR, include your assessment of (a) the business value and (b) the architectural impact (each on a [H,M,L] scale), and briefly outline your reasoning behind each.
- Structure your ASRs in a Utility Tree.

2 QAS Documentation

- Select **three** of the **most highly ranked non-functional requirements** from the utility tree and describe them in further depth and detail in the form of quality attribute scenarios.
- If they are ranked similarly, try to document as different types of quality attribute (diversity).
- Adhere to the template for quality attribute scenarios¹: (i) **Source of stimulus**; (ii) **Stimulus**; (iii) **Environment**; (iv) **Artifact**; (v) **Response**; and (vi) **Response measure**.

¹For more information about the format in which quality attribute scenarios are documented, refer to the book of Bass, Clements and Kazman “Software Architecture in Practice, 3rd ed.”, Part 2 (Quality Attributes). (Available on Toledo)

3 Format of the report

Integrate the results in a coherent document. Adhere to the following structure:

- **A. Utility tree of ASRs.**

Per ASR:

- Summary paragraph (what is the requirement about?). Be short but precise.
- Prioritization (Business value, architectural impact): briefly explain why.

- **B. Quality Attribute Scenarios.** Detailed scenario descriptions.

A $\text{\LaTeX}$ template that imposes the requested structure is available on Toledo. It will help you in documenting these ASRs and QASs in the correct structure. **The template includes a couple example requirements. Do not modify or change them but append your requirements after the examples.**

Submit the PDF report via Toledo. The deadline for this is **11/03**.

Good luck,
The SA team.